

B204A App & Web Development (WS1225)

Website: <https://arpith-binsa.onrender.com>

GitHub Repo: <https://github.com/Arpith-binsa/B204A-App-Web-Development-WS1225->

YouTube Video: <https://www.youtube.com/watch?v=GeH1ClD9E4Q>

Introduction

This Report Includes the Implementation, Deployment, File Structure, Login Credentials and more about a Full-Stack e-commerce web application, Developed for the Final Project for B204 App & Web Development Module.

The Site is built around my personal Photography Hobby, and it allows users to book services like, Sports shoots, Wedding, Automotive, Portrait and Fashion photography. It also allows users to purchase photography gears that I personally use in my field of work.

This project was built using HTML, CSS, and JavaScript for the frontend, Node.js and Express.js for backend, and MongoDB atlas for the database.

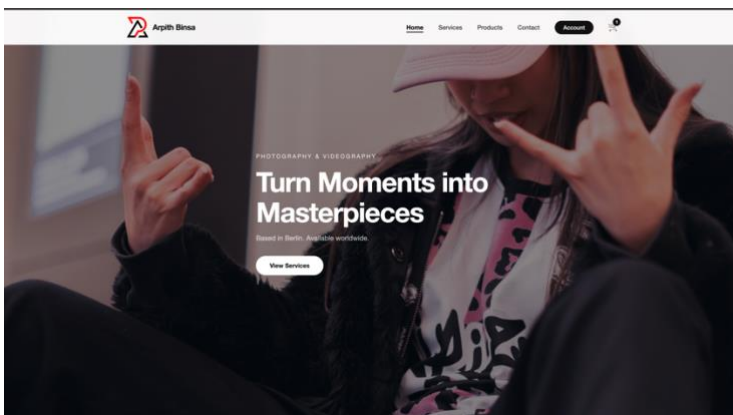
Payments are handles through PayPal Sandbox API. Which Simulates real time transactions without using real bank accounts or balances.

The application is fully functional and deployed at: <https://arpith-binsa.onrender.com>

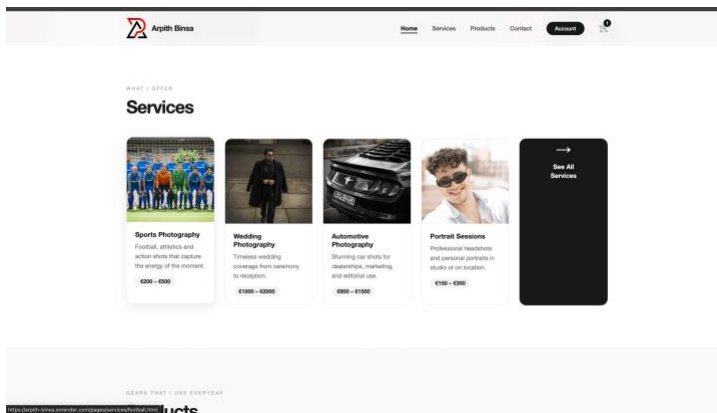
And the complete Source Code is available at: <https://github.com/Arpith-binsa/B204A-App-Web-Development-WS1225->

Website UI

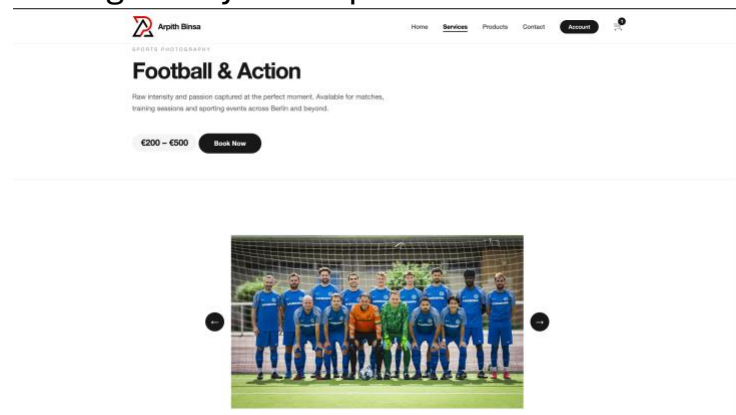
Landing page: The landing page contains a Hero Section with Bold Hero text contrasted with a Large Background image taking up almost the entire screen. This is meant to catch the attention of the user and make the site seem more personal.



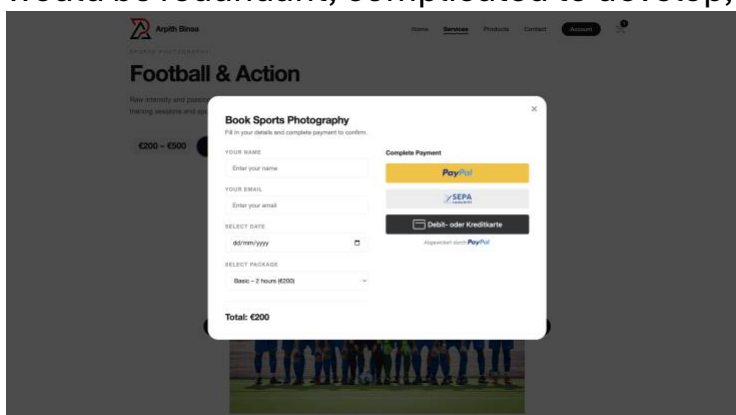
Scrolling down we can see the Service Section and the Products Section with a few products shown at the front which links to the specific product's page.



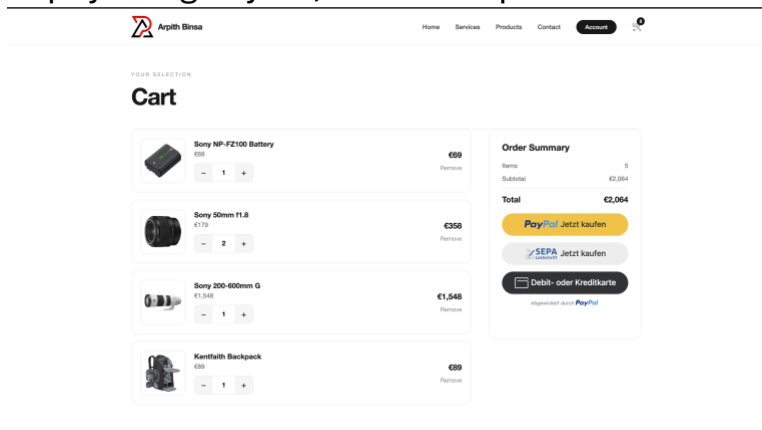
Service pages: Each specific Service page has its own Title, description, price ranges, a “Book Now” button and an image slider with 4 example images to show the users the type of images they can expect.



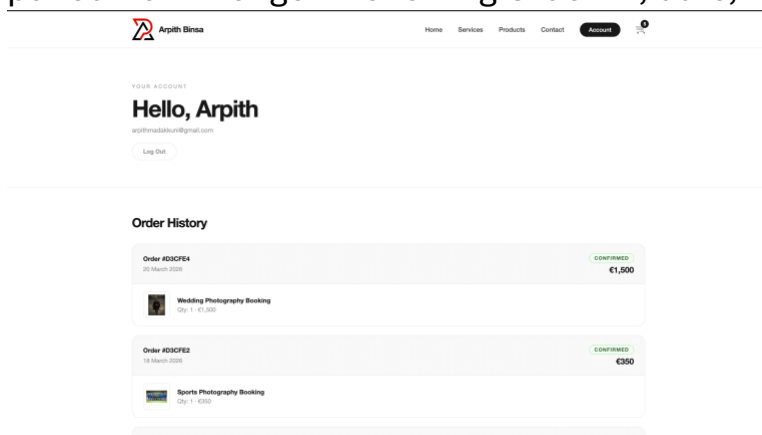
Service Booking Pop-up: When a user clicks the Book now button a pop-up shows up asking for their details and the PayPal script with Buttons for paying using PayPal, Credit Cards or Sepa. It is implemented as a pop-up as individual pages for each service booking would be redundant, complicated to develop, and harder to navigate.



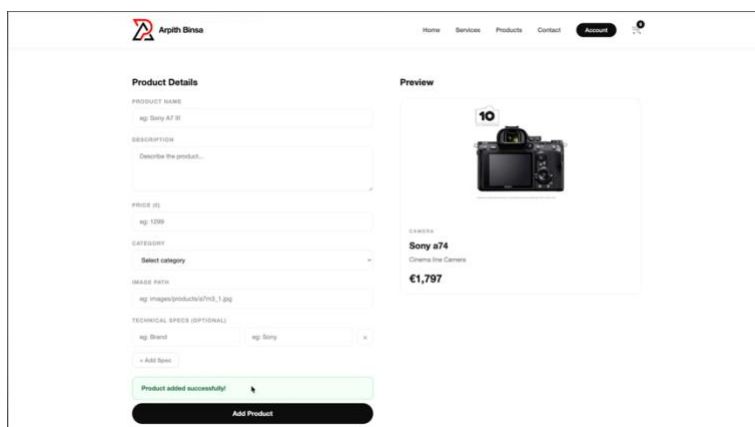
Cart Page: The cart page contains all the items that the user has added to their carts. It calculates the total value of the items and shows summary and a link to PayPal sandbox to pay using PayPal, Card or Sepa.



Account Page: Account page shows you your account email, a button to logout, and if the user isn't logged it, a button to login/register. The user can also view their full order history pulled from MongoDB showing Order ID, date, items and total.



Add Product page: The add product page can be accessed through (arpith-binsa.onrender.com/pages/add-product.html). This feature is exclusive only for admins and cannot be accessed by normal users. This feature allows admins to easily add products. It features a form with fields for Item name, description, price, category, image path and technical specs. When these fields are filled out a preview window is shown, and admins can add products to the Database but



File Structure

The file structure is organized into two areas, the frontend including all the images, HTML files, cart.js and style.css are in the folder named “public”.

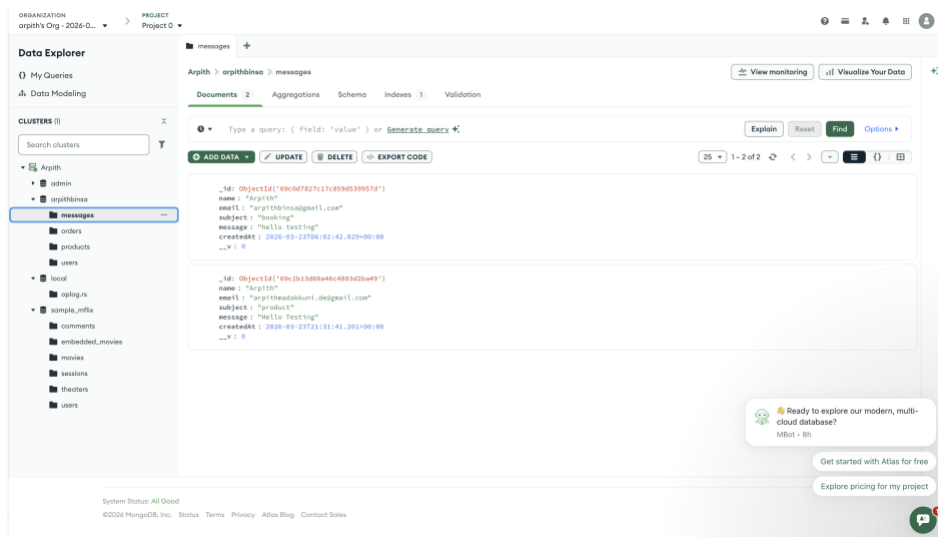
Within the “public” folder both the images and html files for services and products are saved into the images/products, images/services, And the HTML files are saved as pages/products, pages/services.

An exception to this is that index.html, cart.js and style.css are stored in the root folder of “public”.

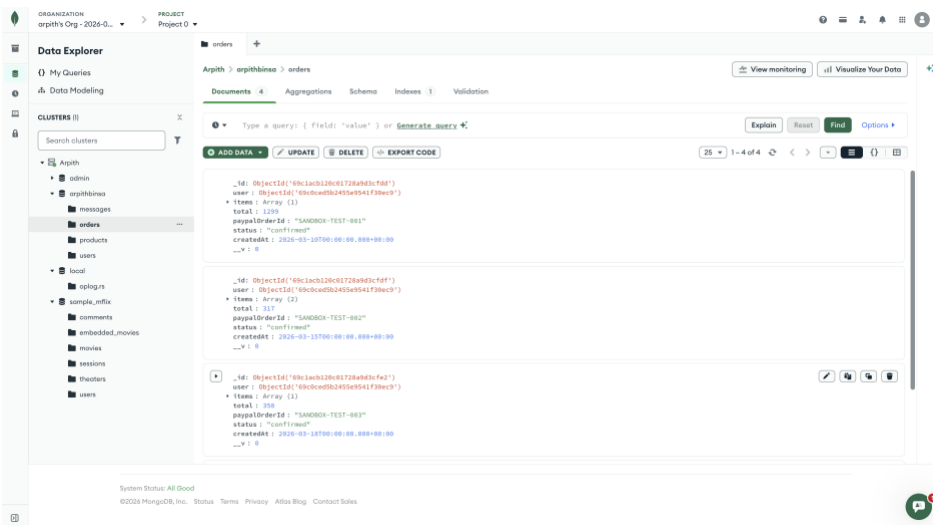
The routes/ folder contains the four route files that handles authentication, products, orders and contact form. The middleware/ folder contains two middleware function, one that verifies JWT tokens and one that restricts certain routes so only admins can use it.

DataBase Schema

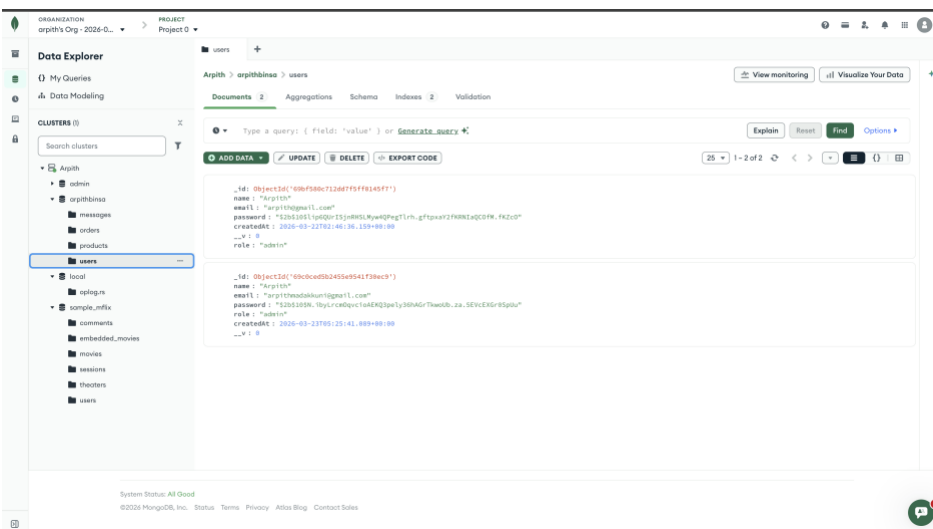
Below are the screenshots for the MongoDB Atlas



Messages sent from the Contact page is saved in the Messages folder.



Order History is stored in the Order folders and called in the frontend code when user checks order history.



All user data is stored in the users folder, users can be made admins either through here or from the vscode terminal.

PayPal Integration

To allow the users to complete the purchase we're using a PayPal Developer tool called PayPal Sandbox. It mimics an actual PayPal account, with an email id, password, PayPal balance, mock credit cards etc.

A PayPal Developer account is needed to create a SandBox environment and get the api keys. Afterwards this API key is used in all the relevant files including Frontend and Backend.

The PayPal button is rendered on two types of pages, the shopping cart page for purchasing all the products in the shopping cart and on the service pages to book a specific service.

When the user clicks the PayPal button the SDK creates an order with all the info like total amount and currency. It then opens up the PayPal popup for the users to log in and complete the purchase.

After the payment the `onApprove` callback is called and the frontend sends the order details to the backend.

```
onApprove: function (data, actions) {  
  return actions.order.capture().then(function (details) {  
    const token = localStorage.getItem('token');  
    const cart = getCart();  
    const items = Object.values(cart);  
    const total = getCartTotal();  
  
    if (token) {  
      fetch('https://arpith-binsa.onrender.com/api/orders', {  
        method: 'POST',  
        headers: {  
          'Content-Type': 'application/json',  
          'Authorization': `Bearer ${token}`  
        },  
        body: JSON.stringify({  
          items: items,  
          total: total,  
          paypalOrderId: details.id  
        })  
      });  
    }  
  });  
}
```

This saves the order to MongoDB and is linked to the logged in user, and it appears on their order history.

Deployment

The Website is hosted on Render.com, a cloud platform that supports full Node.js applications.

It is connected directly to the GitHub Repo. Every time a new commit is pushed to the main branch, Render automatically updates the website and redeploys the application.

Since the website is currently hosted through a free account on Render, the application might take a few minutes to load if it hasn't been loaded in a long time.

A Docker file is included in the Repo which allows anyone to run the application locally without needing to install Node.js.

Updated for the reassessment.

SPA FIX

The previous version of this project was marked down as the frontend wasn't a proper single page application. According to the assessment feedback the site is now changed to a true SPA served entirely from index.html. Using a client-side router.

A routes object maps paths to render functions, navigation uses history.pushstate, the browser's back and forth buttons are handled by a popstate listener, and the server has a catch – all route that returns to index.html so deep links and refreshes work.

The old page which had service and products HTML pages were removed and rebuilt as SPA routes.

Conclusion

Building this project has been a very hands-on and intensive experience, especially writing html, CSS and Js manually instead of using frameworks like React. The initial experience with using API's and tokens has been frustrating yet rewarding.

This project is curated to help with a real-world issue of not having a clear Web page with all the services I provide and the details and prices. The project is built around my personal photography Hobby and could be easily extended into a full website as a practical solution.

Access Credentials

Here are the Credentials needed to fully test the website.

User Email: tester123@gmail.com (admin)

Password: test1234

Live URL: <https://arpith-binsa.onrender.com>

GitHub Repo: <https://github.com/Arpith-binsa/B204A-App-Web-Development-WS1225->

PayPal Sandbox Credentials:

Email: sb-vmunh50114109@personal.example.com (US Based tester Account)

Password: Hc^!S0i}

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